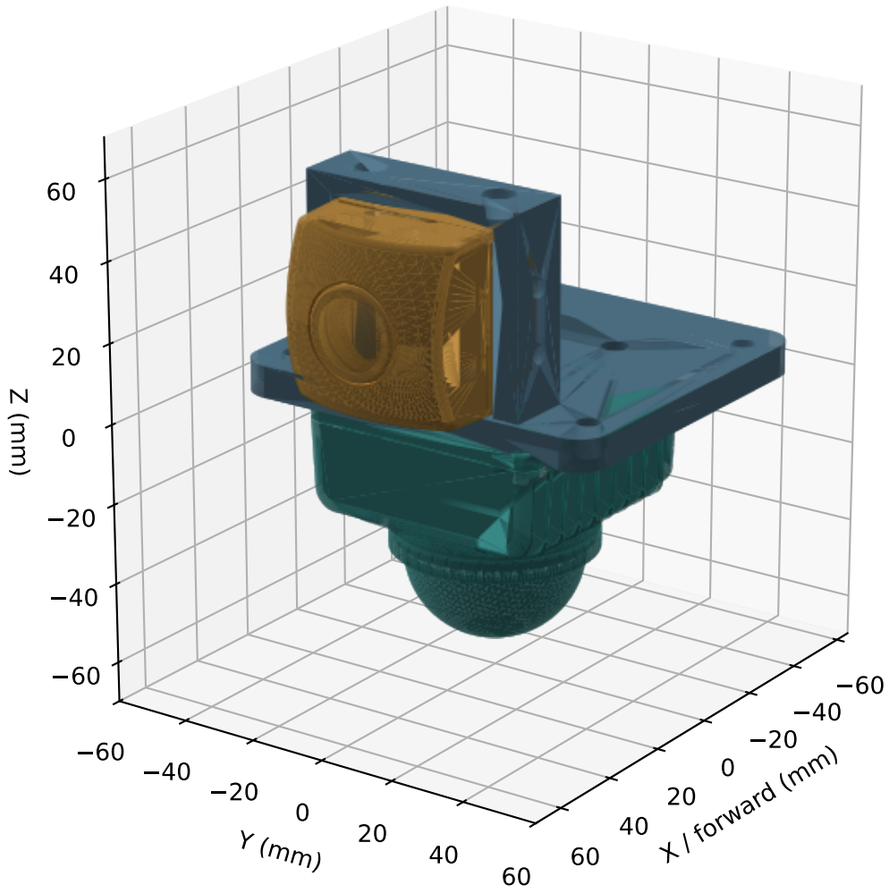
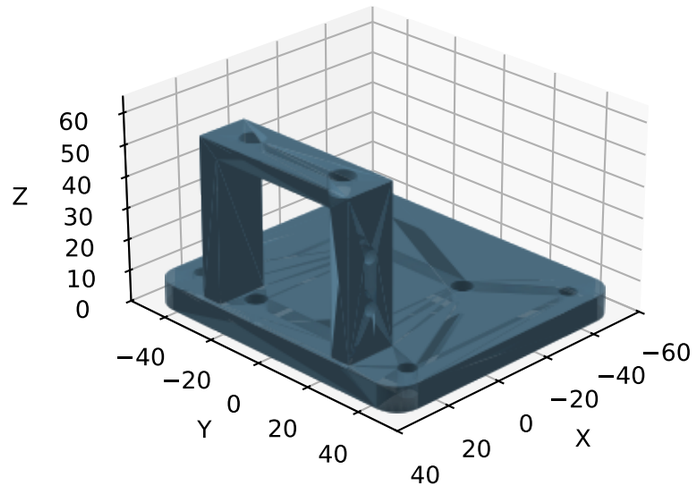
FreeCAD | Single-piece PLA sensor mount

CAMERA +30 mm FORWARD / ONE PRINTED PART / MID-360 DOWN / BOTH CABLE PORTS REARWARD

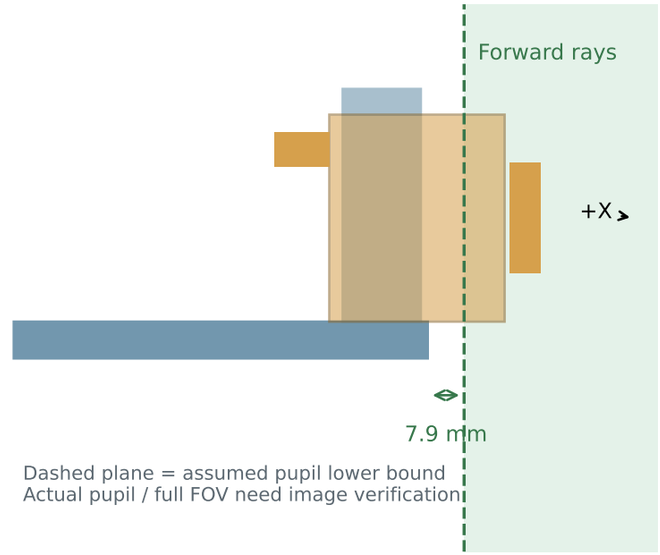
Assembly with supplied sensor STEP models



Print this part only: 90 x 100 x 58.6 mm



SIDE | fixture behind lens barrel (schematic)



PLA FDM: plate underside on bed | supports required inside sleeve | 0.2 mm layers, 5 walls, 40% infill

One printed part + commercial screws/nuts. Verify fit, actual image, clamp retention and temperature before use.

SINGLE-PIECE PLA FIXTURE | Dimensions, printing and assembly (mm)

FreeCAD native feature tree + editable Parameters spreadsheet; exported STL is fixture only.
Base: 90 x 100 x 8, XY center (-10,0), R8.
Sleeve: X=16.5 to 33.5; internal 45.6(Y) x 45.6(Z).
Walls 7; roof 5; overall printed height 58.6.
Camera origin X=30, Y=0; camera bottom Z=8. LiDAR mounting face Z=0, scan -Z.
Camera forward +X. Both cable ports point rearward -X.

BASE HOLES:
 HOST : (-43,-42), dia 5.7 THRU
 HOST : (23,-42), dia 5.7 THRU
 HOST : (-43,42), dia 5.7 THRU
 HOST : (23,42), dia 5.7 THRU
 LIDAR: (-24,-18), dia 3.6 THRU; TOP counterbore dia 6.8 x 3.5 deep
 LIDAR: (-24,18), dia 3.6 THRU; TOP counterbore dia 6.8 x 3.5 deep
 LIDAR: (24,-18), dia 3.6 THRU; TOP counterbore dia 6.8 x 3.5 deep
 LIDAR: (24,18), dia 3.6 THRU; TOP counterbore dia 6.8 x 3.5 deep

Roof driver access: 2 x dia 8 at XY=(24,+/-18), aligned to forward LiDAR screws.
Clamp screw axes: X=25, Z=22.5 and 38.5; direction Y; dia 4.6 through +Y wall.
M4 nut pockets: 7.3 across flats, 3.4 deep from INSIDE of +Y wall.
Use two M4 nylon-tipped screws + two M4 nuts (commercial parts).

PRINT: PLA; 0.4 nozzle, 0.2 layer, 5 walls, 6 top/bottom layers, 40% gyroid, 5 mm brim.
Orient base underside Z=0 on bed. Supports MUST reach sleeve roof from the base top.
Use support on model/everywhere; remove from open front/rear. Deburr holes/pockets.
Temperature/speed follow the actual PLA and printer profile. No machine G-code supplied.

ASSEMBLY:
1. Remove supports. Insert 2 M4 nuts from sleeve interior; back off clamp screw tips.
2. Fasten inverted LiDAR: rear screws directly, front screws through roof access holes.
   M3x8 gives 3.5 mm insertion with the modeled counterbores: verify actual allowed depth.
3. Slide camera in from +X, rear connector first. Seat and tighten nylon tips gently.
   M4x12 is a starting choice; verify actual screw tip/head geometry and reach.
4. Use 4 host M5 holes; host must leave the suspended LiDAR and rear cables clear.

OPTICS: fixture Xmax=35; assumed pupil lower-bound plane X=42.9; rear margin 7.9.
All fixture material is behind this plane, excluding it from the assumed forward hemisphere.
Exact entrance pupil and full calibrated FOV are absent from STEP: verify actual images.
PLA is a prototype material here: verify sensor heating, creep, vibration and retention.
Solid PLA mass at 1.24 g/cm3: 105.4 g; slicer mass differs with infill/support.